II-KN – Completed Cube27 non-splitting, ninth Gauss Fredholm and quantitative reduced resolvent, fourth native Golden H^5, and recurrence dilation-order stratification through 125

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19 August 2026

Abstract

II-KN freezes II-KM under the append-only HT+NT discipline. It upgrades the exact 2-local Cube27 completion to an intrinsic all-stage non-splitting theorem, computes the ninth Gauss transfer trace/Fredholm coefficient and obtains a source-bound quantitative Hardy reduced-resolvent norm, closes a fourth independent native Golden H^5/B^5 carrier (type 5, order 32), and extends the exact recurrence atlas through order 125. The recurrence lane is sharpened beyond reflection by the general polynomial theorem $\operatorname{ord}(f(x^d)) = d\operatorname{ord}(f)$, yielding a primitive-ray divisor law. Cross-singular codimension-three holonomy, the continuous D_5 bundle, the historical $p = 5$ Tate packet, and Stone/Fibonacci project values remain source-gated.

1 Frozen semantics and predecessor

Truth states remain $\{\mathsf{PASS}, \mathsf{ZERO}, \mathsf{SEPARATED}, \mathsf{REFUTED}, \mathsf{NT}\}$; mergeability remains an independent axis. Retrieval, theorem application, external certification, local replay and project-value promotion remain separate evidence channels. II-KM enters at

$$(1457, 100, 460, 295, 90)_{\mathsf{PASS},\mathsf{ZERO},\mathsf{SEPARATED},\mathsf{REFUTED},\mathsf{NT}}, \qquad (687, 464)_{\mathrm{PM},\mathrm{PNM}},$$

with observational continuity `PASS_298` and historical exhaustive replay `NT_NOT_EXHAUSTIVE`.

2 Cube27: exact completion and intrinsic non-splitting

Retain the exact 2-local associated graded from II-KM,

$$\operatorname{gr}^1_{I,(2)} R \simeq (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4, \qquad \operatorname{gr}^2_{I,(2)} R \simeq (\mathbb{Z}/2)^5 \oplus \mathbb{Z}/4,$$

$$\boxed{\operatorname{gr}^n_{I,(2)} R \simeq (\mathbb{Z}/2)^6 \oplus \mathbb{Z}/4 \qquad (n \geq 3).}$$

The normalized augmentation lattices satisfy the exact all-n period-two theorem $K_{m+2} = K_m$ for $m \geq 3$; this remains a theorem on the associated-graded/normalized lattice carrier, not a splitting statement.

The ring $R_{(2)}$ is finite free over $\mathbb{Z}_{(2)}$ and Noetherian. Its I-adic completion $\widehat{R}_{(2)}$ is flat over $R_{(2)}$. Hence it is flat, therefore torsion-free, over $\mathbb{Z}_{(2)}$. For every finite $I^n_{(2)}$,

$$\widehat{I^n_{(2)}} \cong I^n_{(2)} \otimes_{R_{(2)}} \widehat{R}_{(2)},$$

and flatness preserves the injection $I^n_{(2)} \hookrightarrow R_{(2)}$, so each $\widehat{I^n_{(2)}}$ is $\mathbb{Z}_{(2)}$-torsion-free.

Exact completion gives

$$0 \longrightarrow \widehat{I^{n+1}_{(2)}} \longrightarrow \widehat{I^n_{(2)}} \longrightarrow \operatorname{gr}^n_{I,(2)} R \longrightarrow 0.$$

Every displayed quotient has nonzero 2-primary torsion. A $\mathbb{Z}_{(2)}$-linear section would inject that torsion into the torsion-free middle term. Therefore

$$\boxed{\text{the completed filtration sequence is non-split for every } n \geq 1: \ \mathsf{PASS}/\mathsf{REFUTED}(\text{splitting}).}$$

A fortiori there is no continuous $\widehat{R}_{(2)}$-linear section. The exact topological-algebra presentation

$$\boxed{\widehat{R}_{(2)} = \varprojlim_m R_{(2)}/I^m_{(2)}}$$

is retained. A split product of the graded pieces is $\mathsf{REFUTED}$, while an explicit convergent polynomial generators/relations presentation remains $\mathsf{SEPARATED}$. The conductor Ext^1 class is a distinct exact-sequence carrier and is not identified with these filtered extension classes.

3 Gauss: ninth Fredholm and a quantitative reduced resolvent

On the inherited nuclear Gauss carrier the cutoff $N = 6$ gives $6^9 = 10{,}077{,}696$ exact branch words and 235,548 distinct matrix traces. Since nine inverse branches have determinant -1, each fixed-point contribution is $q/(1+q)$ and the positive complement is bounded by the product majorant. At 192-bit Arb precision,

$$\boxed{0.13535937115927398 < \operatorname{tr}(L^9) < 51.806406032125764.}$$

With $\det(I - zL) = 1 + d_1 z + d_2 z^2 + \cdots$, Newton's identity

$$9d_9 = -\big(d_8 \operatorname{tr} L + d_7 \operatorname{tr} L^2 + \cdots + d_1 \operatorname{tr} L^8 + \operatorname{tr} L^9\big)$$

yields

$$\boxed{-6.164604946255728 < d_9 < 4.656333397050539.}$$

The interval crosses zero; no sign is promoted.

For the quantitative lane, Nisoli's certified source data on $H^2(D_1)$ give, on the excluding circle Γ_1 centered at the simple leading eigenvalue 1 with radius 10^{-2},

$$\boxed{\sup_{z\in\Gamma_1} \|(zI - L)^{-1}\| \le 4.25 \times 10^2.}$$

The source repository was frozen at observed main commit `9c0b701f16b41fc94e7dec55f88add183521f915`; the supporting supplementary-data blob is `59f2bb21e3a9d380c85e50549a1a3568f9bf99d5`.

Let P be the leading Riesz projector, $Q = I - P$, and

$$S_H = Q(I - L|_{QH})^{-1}Q.$$

The regular Laurent coefficient satisfies the exact contour identity

$$S_H = \frac{1}{2\pi i}\int_{\Gamma_1} \frac{(zI - L)^{-1}}{z - 1}\,dz.$$

Because $|z - 1| = r$ and $|\Gamma_1| = 2\pi r$,

$$\boxed{\|S_H\|_{H^2(D_1)\to H^2(D_1)} \le 425.}$$

No normality hypothesis is used. This closes the quantitative reduced-resolvent alternative of the II-KN frontier. The inherited AB/BA adapter transfers nonzero spectrum/Riesz data to the weighted branch carrier but is not an isometry; the corresponding weighted operator norm and a proof-grade numerical $P'''(0)$ remain SEPARATED.

4 Golden-Hodge: fourth native H^5/B^5 closure

II-KN uses the actual type-5 stabilizer, not a dimension pattern transferred from another type. Its order is 32 and the native multiplication table contains 29 elementary-abelian E_8 subgroups. The candidate family contains 3884 products built from the actual native H^1, H^2, H^3 representatives.

For each selected E_8, the verifier combines the restricted candidates with an independent actual coboundary block

$$B^5(E) = \operatorname{im}\big(\delta : C^4(E;\mathbb{F}_2) \to C^5(E;\mathbb{F}_2)\big).$$

The joint quotient-rank progression is

$$21 \to 27 \to 33 \to 34 \to \cdots \to 49 \to 53 \to \boxed{56}.$$

After 17 subgroups and

$$\boxed{557056 \text{ streamed } C^5(E_8) \text{ rows,}}$$

the candidate rank reaches the independently known upper bound $\dim H^5 = 56$. Thus

$$\boxed{\dim H^5_{\text{native}}(\text{type } 5) = 56, \qquad \langle\text{displayed native products}\rangle = H^5 : \ \textsf{PASS}.}$$

This is the fourth independent native closure after types $2, 4, 3$. The free $\mathbb{F}_2[H]$-Wall degree-four producer remains SEPARATED; no equivariant free lift is inferred from native/coinvariant bytes.

5 Recurrence: exact orders 116–125 and the dilation theorem

For

$$f_{k,v}(x) = x^k - x^v - 1 \in \mathbb{F}_3[x], \qquad 1 \le v < k,$$

python-flint factorization closes all

$$\boxed{\sum_{k=116}^{125} (k-1) = 1195}$$

new families after a 15/15 frozen-record qualification. The odd orders $117, 119, 121, 123, 125$ have ZERO global-antiperiod sector. The five new fixed centers obey

$$(116, 58) \mapsto (464, 232), \ (118, 59) \mapsto (472, 236), \ldots, \ (124, 62) \mapsto (496, 248),$$

i.e. $\operatorname{ord} C = 4k$ and $h_{\text{anti}} = 2k$. No new family satisfies $\operatorname{ord} C = k$ or $k + v$.

The main refinement is structural. For a monic $f \in \mathbb{F}_p[x]$ with $f(0) \ne 0$, define

$$\operatorname{ord}(f) = \min\{N \ge 1 : f \mid x^N - 1\}.$$

Then for every $d \ge 1$,

$$\boxed{\operatorname{ord}(f(x^d)) = d \operatorname{ord}(f).}$$

Indeed the upper divisibility is immediate. Conversely write $d = p^a e$ with $(e, p) = 1$. If $f(x^d) \mid x^M - 1$, the eth-root-of-unity orbit forces $e \mid M$, while Frobenius substitution multiplies root multiplicities by p^a, forcing $p^a \mid M$. Thus $d \mid M$; writing $M = dL$ and dividing before the injective substitution $y \mapsto x^d$ gives $f \mid y^L - 1$, hence $\operatorname{ord}(f) \mid L$.

Since

$$f_{dk,dv}(x) = f_{k,v}(x^d),$$

one obtains the exact companion-order law

$$\boxed{\operatorname{ord} C(dk, dv) = d \operatorname{ord} C(k, v).}$$

With $g = \gcd(k, v)$,

$$\boxed{\operatorname{ord} C(k, v) = g \operatorname{ord} C(k/g, v/g),}$$

so g always divides the matrix order and $\operatorname{ord} C(k, v)/g$ is a primitive-ray invariant. An auxiliary full atlas $2 \le k \le 125$ certifies 7750/7750 primitive reductions and 12217/12217 divisor scalings with zero violations; this auxiliary scope qualifies the theorem and does not replace earlier carriers.

The main exact atlas $45 \le k \le 125$ contains

$$\boxed{6804 \text{ families}, \quad 1354 \text{ matrix-order collision classes}, \quad 351 \text{ cross-order classes.}}$$

There are 1443 normalized-order multiplicity classes, 898 spanning multiple primitive rays and 1117 primitive rays with multiple dilations in the main window. The equal-order reflection theorem remains compatible with this quotient and passes all 840/840 unordered even-k/odd-v pairs through order 125.

6 Cross-singular and external source gates

Fresh scoped Library/current-input and connected-repository retrieval finds no complete cross-singular codimension-three arrow packet. The alternating C_8 incidence of four S^3 and four T^2 pieces remains a source-bound support shadow; without actual cross-singular arrows, composition/topology and the compactified leaf-groupoid cocycle law remain SEPARATED.

No post-A1BJ/A1BK continuous rank-three D_5 packet is recovered: base, cover, local trivializations, continuous g_{ij}, Cech inverse/triple laws and metric/norm provenance remain NT. The historical $p = 5$ lane still lacks the explicit additive character, self-dual Haar convention, conductor/different denominator, uniformizer/coordinates and finite Tate-sum bytes, so the complete local packet remains SEPARATED. Search zeros are scoped retrieval evidence only.

Stone still lacks project d, θ, K_Z, C and framing/residue provenance; Fibonacci still lacks an instantiated carrier/localization/cofinal-basis/trace-table decisive packet. Therefore

$$\boxed{\text{ABI-complete actual packets} = 0, \quad \text{promoted project values} = 0, \quad 37 \text{ active project gates.}}$$

Parallel Lean/mathlib infrastructure. The previously transferred Lean 4.30.0 fragments were reassembled at archive SHA-256 `1e334ce3...0a611`, extracted into the stable local toolchain, and replayed as Lean 4.30.0 / Lake 5.0.0. All 14486 source-manifest entries were checked bytewise (including the declared symlink-target convention) and the source-line digest reproduces `61dcbcd4...e4d38054`. This is an infrastructure PASS, not a new mathematical proof. No mathlib fragment is mounted at the II-KN freeze; the user-declared no-upstream-manifest mathlib reconstruction is therefore carried to II-KO with local per-fragment SHA creation and archive/TAR validation required before any `import Mathlib` claim.

7 Checkpoint and safe continuation

The append-only checkpoint closes at

$$\boxed{1479\ \mathsf{PASS},\ 102\ \mathsf{ZERO},\ 467\ \mathsf{SEPARATED},\ 297\ \mathsf{REFUTED},\ 93\ \mathsf{NT}},$$

$$\boxed{695\ \mathsf{PROVEN\text{-}MERGEABLE}, \qquad 472\ \mathsf{PROVEN\text{-}NOT\text{-}MERGEABLE}.}$$

The deep verifier closes 12 families with 20,960 explicit assertions, 258,990 witness records and 20,741,896 coverage units. Observational continuity is `PASS_299`; exhaustive historical replay remains `NT_NOT_EXHAUSTIVE`.

Safe successor.

II-KO – Cube27 explicit completed-Rees extension cocycle/pro-system and minimal topological generators-relations or obstruction audit; Gauss tenth Fredholm trace/coefficient plus weighted reduced-resolvent norm transport or proof-grade numerical third-pressure enclosure; codimension-3 cross-singular arrow/topology packet acquisition and compactified leaf-groupoid cocycle realization; Golden fifth native H^5/B^5 closure among types $0, 1, 6$ or trusted free $\mathbb{F}_2[H]$-Wall degree-four producer; post-A1BK/A1BL D_5 continuous transition-source ingestion; historical Pisano $p = 5$ local Tate packet acquisition; recurrence full orders 126–135 and quotient classification modulo reflection/dilation primitive rays; local Lean/mathlib workspace reconstruction and first native formal replay when the transferred archive is complete; plus external Stone/Fibonacci ABI-complete packet execution.

References

[1] II-KM frozen formal bundle, authority outputs and append-only registries.

[2] The Stacks Project, Algebra, Lemma 10.97.2 (tag 00MB), exactness properties of completion for finite modules over Noetherian rings.

[3] I. Nisoli, *Certified spectral approximation of transfer operators and the Gauss map*, arXiv:2602.19435 (2026), with machine-readable certification repository `orkolorko/GKWExperiments.jl`; used as an external Hardy-space resolvent authority only.

[4] D. B. A. Epstein and E. Vogt, *A Counterexample to the Periodic Orbit Conjecture in Codimension 3*, Ann. of Math. 108 (1978), 539–552; inherited source-bound support carrier.

[5] `Nico59000/Modular-recurrence-symetry-analyzer-`; algorithmic recurrence context only, with II-KN high-order factor certification executed locally.